

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Chemical and Volume Control		
TASK:	Perform a Boration Flow Capability to the RCS Surveillance Test		
TASK NUMBER:	N0040430201		
JPM NUMBER:	13-01 NRC Sim a		
ALTERNATE PATH:	☐	K/A NUMBER:	004 A4.18
IMPORTANCE FACTOR:	4.3		4.4
APPLICABILITY:	RO		SRO
EO	☐	RO	☒
STA	☐	SRO	☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	S2.OP-ST.CVC-0011, Rev. 7 (checked 10-7-14)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	8 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	9-6-14
Validated By:	D Tait SME or Instructor	Date:	10-15-14
Approved By:	 Training Department	Date:	11-18-14
Approved By:	 Operations Department	Date:	11-18-14
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

TASK NUMBER: N0040430201

SIMULATOR SETUP: **IC-251**

MODE 3

ET-1

KB127PNB 2CV175 open pushbutton-OPEN

ET-3

KB127TCB 2CV175 close PB(deletes B128)

I/O

OVAO B128 2FI-113A Rapid Borate Flow

Initial 0 Final 33 Ramp 6 seconds

Tied to ET-1

MALF: VL0095 2CV175 fails to 0% open ties to ET-3 with 11 sec TD.

(This prevents reopening valve after it is shut after original IO is deleted when 2CV175 close PB is depressed.

INITIAL CONDITIONS:

Unit 2 is in MODE 3 @ NOP, NOT

INITIATING CUE:

You are the Reactor Operator. Perform S2.OP-ST.CVC-0011, Boration Flow Rate Test

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Complete S2.OP-ST.CVC-0011 and determine 2CV172 test is SAT and 2CV175 test is UNSAT.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide marked up copy of S2.OP-ST.CVC-0011, Boration Flow Rate Test.			
	2.1	IDENTIFY sections of this procedure that are not required to be performed with "N/A".	Determines all sections are to be performed.		
	2.2	REVIEW Components "Off Normal and Off Normal Tagged" List(s) for system and support system(s) associated with evolution to be performed in this procedure.	Determines from initial conditions that Components "Off Normal and Off-Normal Tagged" List(s) review has been performed SAT.		
	2.3	ENSURE one of the following: - The Reactor is in Modes 3-6, or Defueled. <u>OR</u> - The procedure may be performed for Refueling Outages in Modes 1 and 2 <u>provided</u> the rapid boration is started immediately prior to the Reactor Trip.	Determines the Reactor is in Mode 3.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	2.4	ENSURE the Charging System is in service IAW S2.OP-SO.CVC-0001(Q), Charging, Letdown, and Seal Injection.	Determines Charging System is in service.		
	2.5	ENSURE calibration data for the instruments listed in Attachment 1, as applicable, is obtained.	Determines calibration data for the instruments listed in Attachment 1 has been obtained.		
	3.1	Steps identified with dollar signs (\$) are those items required to meet Technical Specification acceptance criteria. Such steps, if not satisfactorily completed, may have reportability requirements and shall be brought to the immediate attention of the SM/CRS.	Reads and initials step.		
	3.2	Due to system limitations, boron addition rate is essentially the same whether one or both Boric Acid Transfer Pumps are operating.	Reads and initials step.		
	3.3	Allow 2CV175, RAPID BORATE STOP VALVE, to full stroke prior to reversing direction.	Reads and initials step.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.4	The ≥ 36 gpm Boric Acid Flow value specified throughout this procedure satisfies the 33 gpm requirement of Technical Specification 4.1.2.2.d plus 3 gpm for instrument uncertainty associated with instrument loops 2FT110 and 2FT113 IAW PR #971016106.	Reads and initials step.		
*	5.1.1	SELECT MAKEUP CONTROL MODE SELECT, STOP.	Selects MAKEUP CONTROL MODE SELECT, STOP.		
*	5.1.2	SELECT 2CV172, BA FLOW CONTROL TO BLENDER, MANUAL.	Selects 2CV172, BA FLOW CONTROL TO BLENDER, MANUAL.		
	5.1.3	OPEN 2CV172.	Determines 2CV172 is open.		
*	5.1.4	On the in service Boric Acid Pump: A. SELECT MANUAL. B. SELECT FAST START.	On the in service Boric Acid Pump selects MANUAL, then FAST START.		
*	5.1.5	SELECT 2CV185, MAKEUP FLOWPATH, MANUAL.	Selects 2CV185, MAKEUP FLOWPATH, MANUAL.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.6	OPEN 2CV185	Opens 2CV185.		
	5.1.7	CLOSE 21CV160, RECIRC VALVE.	Closes 21CV160, RECIRC VALVE.		
	5.1.8	CLOSE 22CV160, RECIRC VALVE.	Closes 22CV160, RECIRC VALVE.		
*	5.1.9	RECORD test results in Attachment 1, Section 2.0.	Initials SAT after verifying 2FI110A indicates ≥ 36 gpm.		
	5.1.10	On the in service Boric Acid Pump, SELECT SLOW START.	Selects SLOW START on in service Boric Acid Pump.		
*	5.1.11	CLOSE 2CV185	Closes 2CV185.		
	5.1.12	PRESS the following RECIRC VLV OPEN (INC FLOW) pushbuttons until the valves are sufficiently OPEN to support BAST level control: A. 21CV160, RECIRC VALVE B. 22CV160, RECIRC VALVE	Throttles open: A. 21CV160, RECIRC VALVE B. 22CV160, RECIRC VALVE		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.13	PLACE the following valves in AUTO: • 2CV185 • 2CV172	Places 2CV185 and 2CV172 in AUTO.		
*	5.2.1	On the in service Boric Acid Pump: C. SELECT MANUAL. D. SELECT FAST START.	On the in service Boric Acid Pump selects MANUAL, then FAST START.		
*	5.2.2	OPEN 2CV175, RAPID BORATE STOP VALVE.	Simulator Operator: Ensure ET-1 is TRUE when 2CV175 OPEN PB is depressed. This inserts the 33 gpm rapid borate flow on 2FI113A. Opens 2CV175, RAPID BORATE STOP VALVE.		
	5.2.3	CLOSE 21CV160, RECIRC VALVE.	Closes 21CV160, RECIRC VALVE.		
	5.2.4	CLOSE 22CV160, RECIRC VALVE.	Closes 22CV160, RECIRC VALVE.		
*	5.2.5	RECORD test results in Attachment 1, Section 3.0.	Initial UNSAT for 2FI113A flow.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Perform a Boration Flow Capability to the RCS Surveillance Test

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2.6	On the in service Boric Acid Pump, SELECT SLOW START.	Selects SLOW START on in service Boric Acid Pump.		
*	5.2.7	CLOSE 2CV175, RAPID BORATE STOP VALVE.	Closes 2CV175, RAPID BORATE STOP VALVE.		
	5.2.8	PRESS the following RECIRC VLV OPEN (INC FLOW) pushbuttons until the valves are sufficiently OPEN to support BAST level control: A. 21CV160, RECIRC VALVE B. 22CV160, RECIRC VALVE	Throttles open: A. 21CV160, RECIRC VALVE B. 22CV160, RECIRC VALVE		
			Terminate JPM when 21 and 22CV160 have been throttled open.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- 0 1. Task description and number, JPM description and number are identified.
- 0 2. Knowledge and Abilities (K/A) references are included.
- 0 3. Performance location specified. (in-plant, control room, or simulator)
- 0 4. Initial setup conditions are identified.
- 0 5. Initiating and terminating Cues are properly identified.
- 0 6. Task standards identified and verified by SME review.
- 0 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 0 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 7 Date 10/15/14
- 0 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: 

Date: 10/15/14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

- Unit 2 is in MODE 3 @ NOP, NOT

INITIATING CUE:

You are the Reactor Operator. Perform S2.OP-ST.CVC-0011, Boration Flow Rate Test